

1. Volatile Oils differ entirely in both chemical and physical properties from fixed oils. They are

A. are capable of volatilization (Distilled from their natural source)

B. are not glyceryl esters of long chain fatty acid (they are terpenoids)

C. Do not become rancid like fixed oils but will oxidize and resinify up on exposure to light and O₂

D. All

2. Why and how are volatile oils present in plants?

A. Waste products: No biological role

B. Detoxification products: Elimination of metabolic products

C. Attraction of insects aiding in pollination

D. Insect repellants defense agents

E. Solvents for wound healing resins

F. All

3. In animals Very few volatile oils are

A. Civet oil from the civet cat

B. Musk oil from the musk deer

C. Ambergris oil from the sperm whale

D. All

4. Volatile Oils

A. Can exist localized,

B. as glycosides

C. Also associated with other groups such as gums and resins

D. All

5. The volatile oil from *Cinnamomum zeylanicum* (Lauraceae):

A. bark is rich in Cinnamaldehyde

B. leaves is rich in Eugenol

C. root is rich in Camphor

D. All

6. Volatile oils are widely used

A. in flavouring (e.g. oil of lemon) and perfumery (e.g. oil of rose)

B. antispasmodic

C. food preservation and in cosmetic preparations

D. as starting materials for the synthesis of other compounds (e.g. oil of turpentine).

E. aromatherapy and as carminatives

F. All

7. For therapeutic purposes volatile oils are administered as

A. inhalations (e.g. eucalyptus oil)

B. orally (e.g. peppermint oil)

C. as gargles and mouthwashes (e.g. thymol)

D. transdermally (many essential oils including those of lavender, rosemary and bergamot are employed in the practice of aromatherapy)

E. All

8. ----- is reactive hemiterpene intermediate in the pathways leading to more complex terpenoid structures.

A. DMAPP B. IPP **C. A and B**

9. Combination the two **DMAPP** and **IPP** via the enzyme **prenyl transferase** yields the monoterpene precursor **geranyl diphosphate (GPP)**

10. **Linalyl PP (LPP)** and **neryl PP (NPP)** are isomers of **GPP** and they are **Acyclic (or linear)** monoterpenes

11. Which of the following is true about cyclic monoterpenes?

A. The bicyclic monoterpenoids are sequentially derived from limonene

B. Because of the *Trans* geometry of the double bond at the C-2 of GPP, direct cyclization to limonene is not possible

C. NPP or LPP, however, do have favourable stereochemistry, immediate precursors of the monocyclic menthane system

D. All

12. Which of the following is monocyclic monoterpenes?

A. Limonene and Carvone

B. beta-phellandrene and alpha-terpineol

C. menthol and menthone **D. All**

13. Which of the following is bicyclic monoterpenes?

A. alpha-pinene and borneol

B. camphor and cineole

C. thujone and thymol

D. All

14. Some aromatic volatile oil components of non-terpenoidal origin (or those derived via the shikimate pathway) and they are

A. cinnamaldehyde B. anethole C. eugenol **D. All**

15. Some components of volatile oils are sesquiterpenes ($C_{15}H_{24}$) and they are

A. cadinene B. zingiberene C. caryophyllene **D. All**

16. Which of the following is true about Glycosides?

A. geraniol, nerol and citronellol occur as glycosides in the petals of *Rosa dilecta*

B. thymol and carvacrol as glucosides and galactosides in *Thymus vulgaris*

C. eugenol, benzyl alcohol, β -phenylethyl alcohol, nerol, geraniol and geranic acid as glucosides in *Melissa officinalis*

D. All

17. All the official volatile oils are extracted by distillation with the exception of

A. oil of lemon B. oil of cade **C. A and B**

18. In which method the plant material is immersed totally in boiling water?

A. Hydro-distillation B. Steam distillation

19. In which method that the heat is difficult to control and hence the rate of distillation is variable and also there is a possibility for overheating and "burning" which lead to a poorer quality oil.

A. Hydro-distillation B. Steam distillation

20. Which methods are that it is rapid, therefore it is less likely to damage those oils?

A. Hydro-distillation **B. Steam distillation**

21. Which method is the most advanced type of distillation?

A. Hydro-distillation **B. Steam distillation**

22. **Rectification** usually takes the form of a second distillation in steam, which frees the oil from resinous and other impurities.

23. *in the enfleurage process*

- A. *glass plates are covered with a thin layer of fixed oil or fat upon which the fresh flowers are spread.*
- B. *The volatile oil gradually passes into the fat*
- C. *The exhausted flowers are removed and replaced by a fresh supply.*
- D. *Formerly the flowers had to be picked off by hand but this is now done mechanically.*

E. All

24. The pneumatic method, which is similar in principles to the enfleurage process, involves the passage of a current of warm air through the flowers.

24. in the digestion process: the flowers are gently heated in melted fat until exhausted, and when they are strained out and the perfume-containing fat is allowed to cool.

-Solvent extraction is based on the **Soxhlet principle**

25. *Isolation of constituents by*

- A. *Fractional distillation*
- B. *Chemical derivatization*
- C. *Gas chromatography*
- D. *Counter-current extraction*
- E. *Crystallization*

F. All

26. *Physical properties of volatile oils*

- A. *Liquid at ambient temperature*
- B. *They have a high refractive index*
- C. *Most of them rotate the plane polarized light.*
- D. *they are soluble enough to impart a distinct fragrance to water /aromatic water)*

E. All

27. *Volatile oils density is lower than water, with the exception*

Clove or cinnamon heavier than water

28. *Chemical properties of volatile oils. They are*

- A. *Complex mixtures*
- C. *Glycosides*
- B. *Terpenes Vs terpenoids*
- D. All**

29. ----- is used to determine the proportions of individual components of certain oils and determination of the chirality of particular components of volatile oils

A. Fractional distillation

B. Chemical derivatization

C. Gas chromatography

D. Counter-current extraction

E. Crystallization

30. Juniper is used for

A. flavouring C. diuretic

B. antiseptic D. aromatherapy **E. All**

31. Lemon is used for

A. aromatherapy C. Perfumery

B. flavouring **D. All**

32. Peppermint is used for

A. flavouring C. Carminative

B. aromatherapy **D. All**

